Reg No.						

SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT TEST – I

Programme: CSE / CSE-Specialization / IT
Course Name: Programming in Python
Course code: S11BLH21
Batch: 2023-2027

Max. Marks: 50
Time: 2 Hours
Year: I
Date: 29-02-24

Part-A **Answer Any Five** (5 ×10 =50)

Q. No.	Questions	CO
1.	Assume the population of a state consists of 'N' persons (both male and female) of different age groups. 1. For the following queries, you are requested to collect the following data: name, date of birth, gender. 2. Identify the number of male and female population in the given state 3. Compute the average age of the gender-based population based on the following group: a) lessthan 5 years b) between 6 and 17 years c) between 18 and 40 years d) between 41 and 60 years e) above 60 years 4. Display the gender-based count of people who are eligible to vote (Use functions)	2
2.	Integers when split into smaller digits, may exhibit the following behaviour a) When split evenly, the resulting pattern will be either in increasing order or decreasing order or unordered b) May not be able to split evenly For e.g 265263 -> 2,6,5,2,6,3 : Even Split - Unordered 265263 -> 26, 52, 63 : Even Split – Increasing Order 265263 -> 265, 263 : Even Split – Decreasing Order 265263 -> Cannot split beyond 3. Prompt the user to input and display the possible combinations Note: Input digits should be in any size	2
3.	An organization is developing a two factor authentication system. The logic varies between the first and the second level authentication. For the first level, it is decided to permit 5 times for entering the passcode and should be blocked thereafter. Similarly for the second level number of times to enter the passcode is 3. a) For First Level: Identify a number (minimum 4 digits) when multiplied by any digit should result in the reverse of the selected number For e.g. 2178 X 4 = 8712 b) For second level: Identify three numbers that are relatively prime to each other and sum them For e.g. GCD(8, 9, 19) = 1 Therefore, Key1: 2178 and Key2: 36	2
4.	Print the PASCAL triangle in the following format (get the number of rows as input) 1 1 2 1 1 3 3 1 1 4 6 4 1 1 5 10 5 1 ...	2
5.	Consider the following dataset:	2

Date	Open	High	Low	Close	Adj Close	Volume
01-02-24	1122.26	1129.68	1097.44	1105.61	1105.61	2343100
02-02-24	1098.26	1126.85	1096.40	1120.83	1120.83	1964900
05-02-24	1119.01	1123.54	1079.81	1097.88	1097.88	2313400
06-02-24	1138.69	1194.66	1130.93	1186.92	1186.92	2664700
07-02-24	1221.45	1225.36	1182.22	1186.51	1186.51	2387300
08-02-24	1206.50	1219.06	1188.16	1210.28	1210.28	1975100
09-02-24	1224.07	1225.56	1196.73	1211.44	1211.44	2175400

For the above data set, plot a bar graph on adj. close, volume, thirty days and forty days simple moving average between any two dates.

6.	Write a Python class labelled as Employee Class Attributes: emp_id, emp_name, emp_salary, and emp_department Class methods and their functions: a) add_employee Method to add a new employee (with all the relevant attributes and the input) b) salary_calculation Arguments: emp_salary, hours_worked If the number of hours worked is over 40, overtime is computed and add it to the salary. overtime = hours_worked - 40 ot_amount = (overtime * (salary / 40)) month_salary = salary + ot_amount c) department_change Method for Changing the employee department d) display Method for displaying the employee details Sample Employee Data: emp_name, emp_id, emp_salary, emp_department Emp1, E8021, 20000, HR Emp2, E8059, 15000, SALES Emp3, E9019, 30000, ACCOUNTS Emp4, E9216, 45000, ADMINISTRATION	2
7.	A system is developed for managing the university's administrative tasks. In this system, the various entities are students, faculty, courses, and departments. Each entity has specific attributes and behaviors, and there are relationships between them. • Define a class Person with attributes name and age. Implement methods to set and get these attributes. • Create a class Student which inherits from Person. Student should have additional attributes such as student_id, courses_taken, and GPA. Implement methods to add courses, calculate GPA, and display student information. • Create a class Faculty which inherits from Person. Faculty should have additional attributes such as employee_id, department, and position. Implement methods to assign courses and display faculty information. • Define a class Course with attributes course_code, course_name, credit_hours, and faculty_assigned. Implement methods to assign faculty to the course and display course information. • Create a class Department with attributes department_name and faculty_list. Implement methods to add faculty to the department, assign courses to faculty, and display department information.	2
8.	Consider the scenario for designing a drawing application. There are various shapes such as circles, rectangles, and triangles. Implement a feature that allows to calculate the total perimeter and total area of all shapes on the canvas. However, it should be able to select specific types of shapes and perform operations only on those shapes. Consider the following structure: 1. Define a base class Shape with abstract methods area() and perimeter(). 2. Create subclasses for different shapes such as Rectangle, Circle, and Triangle, each inheriting from Shape. 3. Implement the abstract methods in each subclass to calculate the area and perimeter specific to that shape. 4. Create a class Canvas to manage a collection of shapes. 5. Implement methods in the Canvas class to add shapes, calculate total area and perimeter, and perform operations on specific types of shapes.	2